

# MediBridge

*World-class care, accessible to all.*

## **Cross-Border Medical Consultations & Surgery Tourism Platform**

Connecting international patients with verified Indian specialists for same-day video consultations and all-inclusive surgery packages.

Client Pitch Document | Confidential

# 1. Executive Summary

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**MediBridge** is an end-to-end digital healthcare platform that bridges the affordability and expertise gap between patients in **Canada, USA & UK** and India's top medical professionals & hospitals.

## The Problem

- Western healthcare costs are 5–10× higher than Indian equivalents for the same quality of care.
- Specialist appointments often require **2–6 weeks** of waiting in North America & the UK.
- International patients lack a trusted channel to verified Indian doctors and JCI-accredited hospitals.
- Medical tourism is fragmented — flights, visas, hospitals, and follow-up are all handled separately.

## Our Solution

A single, secure web platform offering **two integrated products**:

1. **Online Consultations** — Same-day video appointments with verified Indian doctors, with digital prescriptions delivered as signed PDFs.
2. **Surgery Tourism Packages** — All-inclusive bundled packages (surgery + flight + hotel + visa + transfers) at partner hospitals, with end-to-end coordination via the admin team.

## Why Now?

- **Telemedicine market** projected to reach \$460B by 2030 (CAGR 24%).
- **Medical tourism to India** projected at \$13B by 2026.
- Post-pandemic comfort with video consultations is now mainstream.

## 2. Platform Snapshot

Current build — production-ready MVP.

|                            |                         |                   |                         |
|----------------------------|-------------------------|-------------------|-------------------------|
| 25                         | 22                      | 8                 | 34                      |
| Verified Doctors Onboarded | Medical Specializations | Partner Hospitals | Active Surgery Packages |
| 154                        | 35                      | 4                 | 12                      |
| Backend Tests Passing      | Frontend Pages          | User Roles        | Build Phases Complete   |

### Platform Highlights

- **100% test coverage** across all critical flows (auth, booking, prescriptions, surgery, admin).
- **Mobile-first responsive UI** — works on phone, tablet, and desktop with adaptive layouts.
- **Dark mode** support across all 35 pages.
- **JCI-accredited hospital network** across India's top metros.
- **14+ surgery categories** — Cardiac, Neuro, Orthopedic, Oncology, Transplant, etc.

## 3. User Roles & Privileges

MediBridge supports **4 distinct user roles**, each with carefully scoped privileges enforced both at the API layer (Django permissions) and the UI layer (RoleGuard component).

### Patient Primary User

|                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------|
| What Patients Can Do                                                                                                                               |
| <ul style="list-style-type: none"><li>• Create an account with email verification.</li></ul>                                                       |
| <ul style="list-style-type: none"><li>• Manage personal profile (DOB, gender, address, medical history, allergies, current medications).</li></ul> |
| <ul style="list-style-type: none"><li>• Upload medical reports and historical documents for doctor review.</li></ul>                               |
| <ul style="list-style-type: none"><li>• Browse the verified-doctor directory, filter by specialization, view profiles.</li></ul>                   |

- Submit symptom intake forms describing health concerns.
- Book consultation slots based on real-time doctor availability.
- Receive Jitsi video call links 15 minutes before scheduled appointments.
- Cancel scheduled appointments (cannot cancel completed appointments).
- View digital prescriptions with medicines, tests, diagnosis — download as signed PDF.
- Confirm or decline doctor-proposed follow-up appointments.
- Browse surgery package catalog with filters by procedure.
- Submit a multi-step surgery booking (package → travel info → documents → payment).
- Upload passport, visa, insurance, and other travel documents securely.
- Track surgery booking status from *info-pending* through *confirmed*.
- Download surgery voucher PDF with QR code after confirmation.

### What Patients Cannot Do

- View other patients' data, appointments, or documents.
- Modify a prescription after the doctor has issued it.
- Book a confirmed surgery directly without admin approval of any surgery recommendation.
- Access doctor or admin endpoints (enforced via role permissions).

## Doctor [Medical Professional](#)

### Onboarding Flow

- Receives admin invitation via email (token-gated, 7-day expiry).
- Completes profile during signup — bio, education, license, fee, languages.
- Admin verifies credentials before doctor becomes publicly visible.

## What Doctors Can Do

- Set weekly recurring availability slots (e.g., Mon-Fri 9am-5pm IST).
- Override with specific-date slots (e.g., extra Saturday clinic).
- Delete future availability slots.
- Manage profile — bio, hospital affiliation, languages, consultation duration, fee.
- Add multiple education entries (degree, institution, year).
- View list of own scheduled, in-progress, completed, and no-show appointments.
- Read patient's full medical profile (history, allergies, reports) before consultation.
- Transition appointment status: scheduled → in-progress → completed / no-show.
- Write digital prescription — medicines (dosage, timing, duration), diagnostic tests, diagnosis, notes.
- Edit prescription within a 24-hour window after issue.
- Propose follow-up consultations to the patient (fee-waived or fee-paid).
- Recommend a surgery package to the patient (sends to admin for approval).
- View own surgery recommendations and their approval status.

## What Doctors Cannot Do

- Self-verify (must be admin-verified before going live).
- Access other doctors' patients, appointments, or prescriptions.
- Edit a prescription after 24 hours.
- Directly create surgery bookings (recommendations go through admin).
- Access admin endpoints (intakes, audit log, hospitals, packages).

What Admins Can Do

- **Operations Dashboard** — live KPIs: pending intakes, today's appointments, new surgery bookings, unverified doctors, active doctors, surgery revenue, consultation revenue.
- **Symptom Intake Triage** — review patient symptom submissions and match to appropriate doctor.
- **Doctor Onboarding** — invite new doctors by email, set their specialization, review credentials, verify or unverify accounts.
- **User Management** — list all users, edit user fields, reset passwords, activate / deactivate.
- **Hospital Management** — CRUD operations on partner hospitals (name, city, state, accreditations, website, description).
- **Surgery Package Management** — create / edit packages with pricing, duration, inclusions, exclusions, partner hospital linkage.
- **Booking Oversight** — view all consultations and surgery bookings across the platform, filter by type and status.
- **Surgery Booking Workflow** — review travel info, validate documents, confirm bookings which triggers automatic voucher generation.
- **Surgery Recommendation Approval** — review doctor-recommended packages, approve or reject with notes before patient can book.
- **Audit Log** — immutable record of every state-changing action (verifications, confirmations, matches) with actor + timestamp.

What Admins Cannot Do

- Impersonate users or read messages between patient and doctor.
- Modify a prescription (write-protected for medical & legal reasons).
- Bypass audit logging — every admin action is recorded.

What Visitors Can See (No Login Required)

- Marketing home page with hero, specialty grid, testimonials, and CTA.

- Public doctor directory — verified doctors only, filterable by specialization.

- Individual doctor profiles — bio, education, languages, fee, available slots.

- Public package catalog — price, duration, inclusions, hospital partner.

- Individual package detail pages with full breakdown.

- Authentication pages — sign up, log in, forgot password, reset password.

## What Visitors Cannot See

- Any patient or doctor private data.

- Unverified doctors (admin-gated).

- Inactive surgery packages.

- Any internal admin or operations data.



## 4. Core Features (Phase-by-Phase)

Delivered across 12 sequential implementation phases — all complete.

| # | Phase                    | Key Capabilities Delivered                                                                                                        |
|---|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 0 | Skeleton & Environment   | Project scaffold, MySQL setup, PyMySQL shim, dev/prod settings split                                                              |
| 1 | Authentication           | JWT in httpOnly cookies, email verification (24h tokens), password reset (1h tokens), bcrypt password hashing                     |
| 2 | Patient Profile          | DOB, gender, address, medical history, allergies, current medications, medical reports upload                                     |
| 3 | Doctor Onboarding        | Admin-invite → token-gated signup → profile completion → admin verification → public listing                                      |
| 4 | Availability & Directory | Recurring + specific-date slots, timezone-aware UTC storage, public doctor directory, specialization filter                       |
| 5 | Symptom Intake           | Patient submits symptoms, admin reviews and matches to best-fit doctor, optional matching notes                                   |
| 6 | Booking & Payments       | Slot selection, concurrency-safe with <code>select_for_update</code> , dummy payment endpoint (DEBUG only), Jitsi link generation |
| 7 | Digital Prescriptions    | Medicines with dosage/timing JSON, diagnostic tests, diagnosis text, xhtml2pdf-rendered PDF, 24h doctor edit window               |

|    |                         |                                                                                                           |
|----|-------------------------|-----------------------------------------------------------------------------------------------------------|
| 8  | Follow-up Consultations | Doctor proposes follow-up (fee-waived or fee-paid), patient confirms, integrates with payment flow        |
| 9  | Hospitals & Packages    | Hospital CRUD with accreditations, package CRUD with inclusions/exclusions, slug-based public URLs        |
| 10 | Surgery Booking         | 4-step wizard (package → travel info → documents → payment), passport upload, voucher PDF with QR code    |
| 11 | Admin Panel             | KPI dashboard, audit log (immutable), user management, booking oversight, surgery recommendation approval |
| 12 | Polish & Handover       | Mobile responsiveness, dark mode, error states, loading states, SEO meta, sitemap, robots.txt             |

## 5. Use Cases

### Use Case 1: International Patient Books an Online Consultation

1. Patient (Toronto) signs up at MediBridge, verifies email, completes profile.
2. Submits symptom intake describing persistent chest discomfort.
3. Admin reviews the intake and matches the patient to **Dr. Sharma (Cardiologist)**.
4. Patient receives email notification, browses Dr. Sharma's available slots in IST.
5. Books 4pm IST slot (12:30am ET) — dummy payment flow generates payment reference.
6. 15 min before appointment — both receive Jitsi video link.
7. Doctor reviews patient profile, conducts video consult, marks "in progress" → "completed".
8. Doctor issues prescription: aspirin 75mg, lipid panel test, ECG, follow-up in 7 days.
9. Patient downloads signed PDF prescription; confirms doctor-proposed follow-up.

### Use Case 2: Patient Books a Bundled Surgery Package

1. Following the consultation, doctor recommends a **bariatric surgery package** at Apollo Chennai.
2. Admin reviews and approves the recommendation (with notes).
3. Patient sees approved recommendation, opens the booking wizard.
4. Reviews package: \$9,500 USD inclusive of 5-day hospital stay, 7-day recovery hotel, flights, transfers, visa support.
5. Enters travel info — passport number, country, expiry, current occupation.
6. Uploads passport scan + insurance certificate (stored in authenticated-access folder, NOT public MEDIA\_URL).
7. Completes dummy payment.
8. Admin validates docs, confirms booking — **voucher PDF with QR code auto-generated**.
9. Patient downloads voucher, presents at hospital reception via QR scan.

### Use Case 3: Admin Onboards a New Specialist

1. Admin clicks "Invite Doctor" on admin panel.
2. Enters email, name, primary specialization — 7-day signup token emailed.
3. Doctor receives email, clicks link, sets password, fills in bio, license, fee, languages, education.
4. Profile shows "Pending Verification" badge — not visible in public directory.
5. Admin reviews credentials, marks *verified* — audit log records the action.
6. Doctor immediately appears in public directory with verified badge; can accept bookings.

## 6. Tools & Technologies

### Backend Stack

| Layer          | Technology                        | Why We Chose It                                                     |
|----------------|-----------------------------------|---------------------------------------------------------------------|
| Web framework  | Django 6.0                        | Mature, batteries-included, security-first                          |
| API layer      | Django REST Framework 3.17        | Industry-standard for Python APIs, integrates with permissions/auth |
| Authentication | djangorestframework-simplejwt 5.5 | JWT tokens, kept in httpOnly cookies (XSS-safe)                     |
| Database       | MySQL 8.0+                        | Client requirement; production-tested with PyMySQL driver           |
| API docs       | drf-spectacular 0.29              | Auto-generated OpenAPI 3.0 + Swagger UI                             |
| PDF generation | xhtml2pdf 0.2.17                  | Pure-Python, Windows-friendly, HTML/CSS templates                   |
| QR codes       | qrcode 8.2                        | For surgery voucher QR codes                                        |
| Images         | Pillow 12.2                       | Avatar resizing, document thumbnail                                 |
| CORS           | django-cors-headers 4.9           | Locked-down origin policy for FE-BE separation                      |
| Config         | python-decouple 3.8               | 12-factor app envvar handling                                       |
| Testing        | pytest 9 + pytest-django 4.12     | 154 tests, factory-boy fixtures                                     |

### Frontend Stack

| Layer     | Technology              | Why We Chose It                                      |
|-----------|-------------------------|------------------------------------------------------|
| Framework | Next.js 16 (App Router) | Server components, file-based routing, SSR/ISR ready |

|                      |                          |                                                     |
|----------------------|--------------------------|-----------------------------------------------------|
| Language             | TypeScript 5 (strict)    | Compile-time type safety, fewer runtime bugs        |
| UI library           | React 19                 | Latest stable; concurrent rendering                 |
| Styling              | Tailwind CSS 4           | Utility-first; fast iteration; built-in dark mode   |
| Component primitives | shadcn/ui + Base UI      | Accessible, unstyled primitives we own (no lock-in) |
| Server state         | TanStack Query 5         | Caching, deduplication, optimistic updates          |
| Forms                | React Hook Form + Zod    | Type-safe validation, minimal re-renders            |
| HTTP client          | axios 1.16               | Cookie support, interceptors for auth refresh       |
| Date / TZ            | date-fns 4 + date-fns-tz | Lightweight, tree-shakeable; cross-timezone display |
| File upload          | react-dropzone 15        | Accessible drag-drop with file-type guards          |
| Notifications        | sonner 2                 | Native-feeling toast notifications                  |
| Theming              | next-themes 0.4          | System / light / dark mode toggle with no FOUC      |
| Icons                | lucide-react 1.14        | Tree-shakeable SVG icon set                         |

## Architecture Decisions

- Authentication boundary:** JWTs stored in `HttpOnly` cookies with `SameSite=Lax` — XSS-resistant. **Not** stored in `localStorage`.
- Sensitive uploads:** Passports, insurance, and medical documents are stored outside `MEDIA_URL` and served only through authenticated views with per-request ownership checks.
- Booking concurrency:** Slot reservations use `select_for_update()` inside a DB transaction to prevent double-booking under concurrent load.

**Timezones:** All timestamps stored as UTC; the frontend converts to the patient's local timezone using `date-fns-tz`.

**Dummy payments:** The dev payment endpoint refuses to run when `DEBUG=False` — production hookup planned for Stripe / Razorpay.

## 7. Security & Compliance

### Authentication & Authorization

- Passwords hashed with Django's PBKDF2 (10,000 iterations).
- JWTs in `HttpOnly`, `Secure`, `SameSite=Lax` cookies.
- Email verification mandatory before login.
- Password reset tokens expire in 1 hour, single-use.
- Doctor signup tokens expire in 7 days, single-use.
- Per-endpoint permission classes: `IsAdmin`, `IsDoctor`, `IsPatient`, `IsAuthenticated`.

### Data Protection

- Cross-patient isolation enforced at the queryset level (not just UI hiding).
- Cross-doctor isolation: doctors can only see/edit their own appointments and prescriptions.
- Sensitive files (passports, medical reports) served via authenticated views with ownership checks.
- CORS policy whitelisted to known frontend origin only.
- Audit log records every state-changing admin action with timestamp + actor.

### Input Validation

- All requests validated by DRF serializers before hitting business logic.
- File uploads checked for: MIME type, max size (10 MB), allowed extensions.
- Frontend forms double-validate with Zod schemas for instant feedback.
- SQL injection prevention via Django ORM (parameterized queries).
- XSS prevention via React's escaping + `httpOnly` cookies.

## 8. Quality Assurance

### Test Coverage (154/154 passing)

| Test Suite        | Tests | Coverage                                                      |
|-------------------|-------|---------------------------------------------------------------|
| Authentication    | 14    | Signup, login, verify, reset, logout, expiry                  |
| Patient Profile   | 8     | Auto-create, get/patch, isolation, DOB validation             |
| Doctor Onboarding | 17    | Invite, token signup, verification, education CRUD, isolation |

|                          |    |                                                               |
|--------------------------|----|---------------------------------------------------------------|
| Availability & Directory | 15 | Slot CRUD, recurring expansion, TZ conversion, public listing |
| Symptom Intake           | 17 | Submit, cancel, admin match, audit log                        |
| Appointments             | 13 | Book, cancel, status transitions, double-booking prevention   |
| Prescriptions            | 12 | Create, edit-window enforcement, PDF generation, isolation    |
| Follow-up                | 9  | Propose, confirm, fee-waived/paid, slot conflict              |
| Hospitals & Packages     | 16 | CRUD, public listing, slug generation, active filter          |
| Surgery Booking          | 17 | Create, travel info, document upload/delete, confirm, voucher |
| Admin Panel              | 10 | Dashboard KPIs, bookings, audit log, permission gates         |
| Health Check             | 1  | Liveness endpoint                                             |

## Frontend Quality

- TypeScript strict mode — 0 type errors across 35 pages.
- Mobile-first responsive design tested at 375 / 768 / 1024 / 1440 px.
- Dark mode parity on every page.
- Loading skeletons on every async fetch.
- Error toasts via Sonner for every failed API call.
- SEO — meta tags, sitemap.xml, robots.txt, semantic HTML.



## 9. Roadmap (Post-MVP)

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### Phase A: Payment & Notifications (Next Sprint)

- Stripe / Razorpay integration replacing dummy payment endpoint.
- Refund & partial-refund flows for cancellations.
- Email + SMS notifications for booking confirmations, reminders.
- WhatsApp Business API integration for patient outreach.

### Phase B: Live Video & Telephony

- Migrate from Jitsi to **Twilio Programmable Video** for HIPAA-compliant calls.
- In-app chat between patient and doctor (with file attachments).
- Voice-only fallback for low-bandwidth patients.
- Recording with patient consent + secure S3 storage.

### Phase C: AI & Triage

- AI-assisted symptom-to-specialty matching using LLM (Claude / GPT).
- OCR for medical-report extraction at upload time.
- Automated prescription suggestions from doctor's prior patterns.
- Multi-language support (Hindi, Arabic, Spanish, French).

### Phase D: Native Mobile

- React Native app (iOS + Android) sharing the existing TypeScript types.
- Push notifications for appointment reminders.
- Biometric login (Face ID, Touch ID).
- Offline-first patient profile + prescription viewing.

### Phase E: Compliance & Scale

- HIPAA compliance audit + Business Associate Agreement with cloud provider.
- SOC 2 Type II certification.
- Multi-region deployment (US-East + Mumbai) for low-latency video.
- EMR integrations with Apollo, Fortis, Max hospital chains.

## 10. Why MediBridge Wins

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| Pillar | What It Means For The Client |
|--------|------------------------------|
|--------|------------------------------|

|                            |                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------|
| Two products in one        | Cross-sell from consultation to surgery package — higher LTV per patient.                                 |
| Verified-only network      | Trust signal for Western patients hesitant about offshore care.                                           |
| Built-in admin tooling     | Operations team can onboard doctors, verify packages, and confirm surgeries with no engineering overhead. |
| Mobile-ready from day one  | ~60% of healthcare research starts on mobile — we don't lose those visitors.                              |
| Type-safe, tested codebase | Fast feature iteration without regressions — tests are the safety net.                                    |
| Standards-based stack      | Easy hiring — Django and Next.js have huge talent pools.                                                  |

## 11. Next Steps

1. **Live demo** — we walk through patient signup → consultation → prescription → surgery booking in real time on desktop and mobile.
2. **Custom branding** — replace MediBridge logo & color palette with client brand.
3. **Production deployment** — provision MySQL, set up domain, SSL, CI/CD pipeline.
4. **Real payment gateway** — integrate Stripe or Razorpay (live keys).
5. **Doctor onboarding sprint** — bulk-invite the client's existing doctor network.
6. **Hospital partnerships** — add the client's preferred hospitals + packages.
7. **Soft launch** with 50 beta patients across CA / US / UK.